

ADCHARAN ARIVUCHELVAN

☎ 647-467-9552 ✉ aarivuch@uwaterloo.ca [in linkedin.com/adcharan](https://www.linkedin.com/adcharan) [Portfolio Site](#)

Education

University of Waterloo

Sep. 2023 – Apr. 2028

Bachelor of Applied Science in Mechanical Engineering

Waterloo, ON

Combined GPA: 3.75

Relevant Coursework: Digital Computation, Electrical Engineering, Intro to Mechanical Engineering, Mechanics, Material Structure, Electromechanical Devices, Digital Logic and PLC Programming, Thermodynamics

Skills

Languages: Java, C++, C, HTML/CSS, Python, Javascript, MATLAB, PLC Ladder Programming

Tools: 3D Printing, AutoCAD, Solidworks, Fusion 360, Technical Drawing, MiniTab, FEA, GD&T, Office, Machining

Certifications: CSWP, CSWA, Azure Fundamentals - AZ 900, Azure AI Fundamentals - AI 900

Experience

Linamar Corporation - Camcor Manufacturing

Jan 2025 – April 2025

Manufacturing Engineering Co-Op

Guelph, ON

- Worked with Camcor Team to launch new production line of a rear drive module and power transfer unit in the KM Jeep Cherokee. Solved manufacturing issues in quality, fixtures, documentation and safety.
- Designed a conveyor cart using pre-designed cart assemblies on **SolidWorks** to lift and hold rejected part pallets from the production line. Used **SolidWorks** to create Pinion Fixtures, Part Disassembly Tools and Oil Soak Tools.
- Created blueprints for tools to be machined in tool room and process specifications based on **FMEA & PFMEA** using **SolidWorks Drawings**. Edited and created production line layouts using **AutoCAD** and **DraftSight**.
- Collected data for quality analysis and gauge calibration of machines by running parts, creating probability distributions on part usage, and GRR reports on **MiniTab**. Created documentation for line using **Excel** and **Word** to create Bill of Materials, LOTO placards, safety alerts, worker instruction visual aids, CMM blueprints etc.

WE Accelerate Microsoft Azure A.I

May 2024 – Aug 2024

Intern

Waterloo, ON

- Developed a machine learning pipeline prototype using **Azure Machine Learning Designer** which can predict housing sale prices with more precision compared to top real estate listing sites such as Zillow.
- Leveraged **Azure Cloud and AI** technologies to innovate in the real estate sector by developing intelligent systems that enhance buyer decision-making, optimize bid strategies, and ensure more accurate property valuations.
- Collaborated with machine learning experts at Avanade to develop an AI solution using **Microsoft Azure** tools that optimize bidding strategies by alerting users to potential bidding wars.

Midnight Sun Solar Race Car Team - Electrical/Dynamics Team Member

Sept 2023 - May 2025

Student Design Team

Waterloo, ON

- Helped to design a hydraulic braking system with potential for regenerative charging with **SolidWorks**. Designed and constructed PCBs using **Altium Designer** for various subsections such as battery management, power distribution and solar sense. Built a practical, easy-to-debug, and **resilient electrical system**, taking full advantage of both hardware and software capabilities. Machined linkages and shafts for the vehicle.
- Helped design a jig in **Solidworks**, and ran **FEA** tests on parts and weldments to determine design feasibility.

Projects

Self Driving RC Car | *Python, Raspberry Pi, SolidWorks, AutoDesk*

August 2024

- * Developed a self-driving RC car using a machine learning algorithm implemented on a **Raspberry Pi Zero 2W**, equipped with a 30 FPS Raspberry Pi camera for real-time navigation. Developed an image classification program using the **PyTorch** library to predict the correct trajectory with 90 percent accuracy.
- * 3D printed and utilized **SolidWorks** and **Autodesk Inventor** to design the **Rack and Pinion** steering system, a **Double Wishbone Suspension** and a durable frame. Appropriately wired the motors for the back wheel drive and the motor for the steering with the motor controller and the **Raspberry Pi**.

Arduino Steering Wheel | *Python, SolidWorks, Arduino*

February 2025

- * Created a program using an accelerometer, gyro sensor and **Python** to register the turning of the steering wheel to map to A and D keys for steering. Also coded 2 pressure sensor to act as pedals for W and D keys.
- * Designed a steering wheel hub and mount using **SolidWorks** to house Arduino, breadboard and sensors along with a wooden latch mechanism for the pedals.

Romanoake | *Python, HTML/CSS, Javascript*

July 2024

- * Developed a web karaoke application leveraging the **Spotify API** and **Azure Large Language Models (LLMs)** to transliterate LRC files into the Latin alphabet, enhancing accessibility for users.
- * Designed a scoring system for a web karaoke app that evaluates users based on the artist singing each lyric, utilizing Spotify API's music analysis feature to group timbre vectors by similarity, distinguishing between human vocals and instruments to score user performances accurately.